

SIEM Visualization Lab – Failed Logon Attempts

Dashboard

This lab focuses on building a Security Information and Event Management (SIEM) dashboard using **Kibana**. The objective was to create and refine a dashboard that visualizes failed logon attempts across Windows systems, helping SOC analysts identify suspicious authentication activity.

Objectives:

- Create a new SIEM dashboard from scratch
- Build a visualization for failed Windows logon attempts
- Apply filters using event IDs (4625)
- Aggregate and analyze user and host data
- Refine visualization based on SOC requirements

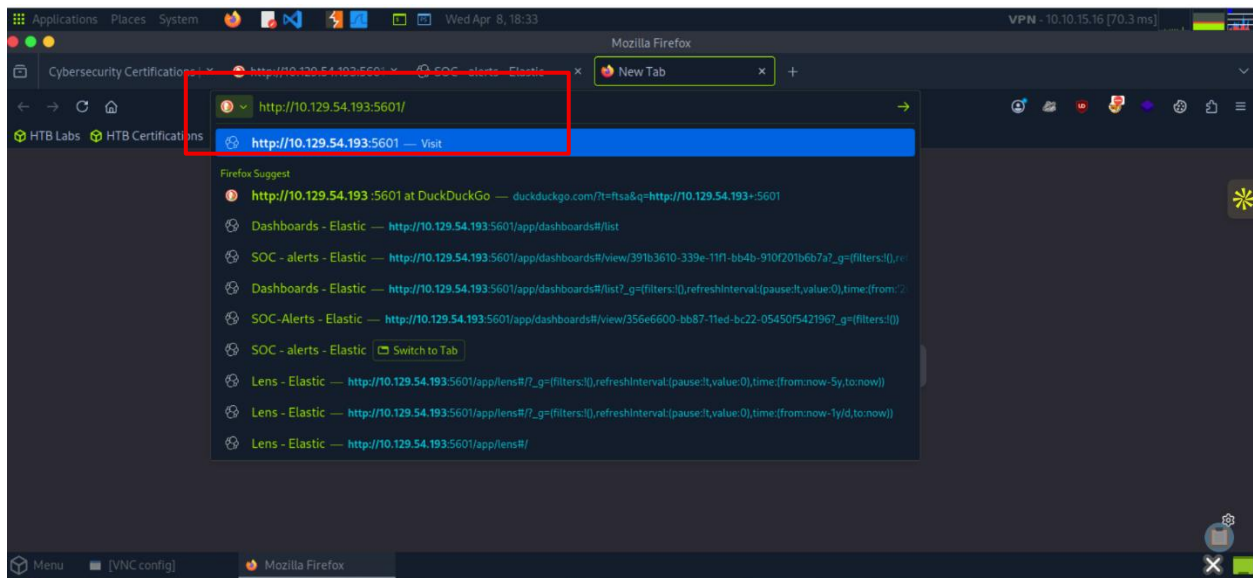
Tools & Technologies:

- **Kibana** – Data visualization and dashboard creation
- **Elasticsearch** – Data storage and indexing
- Windows Event Logs (Security logs)
- KQL (Kibana Query Language)

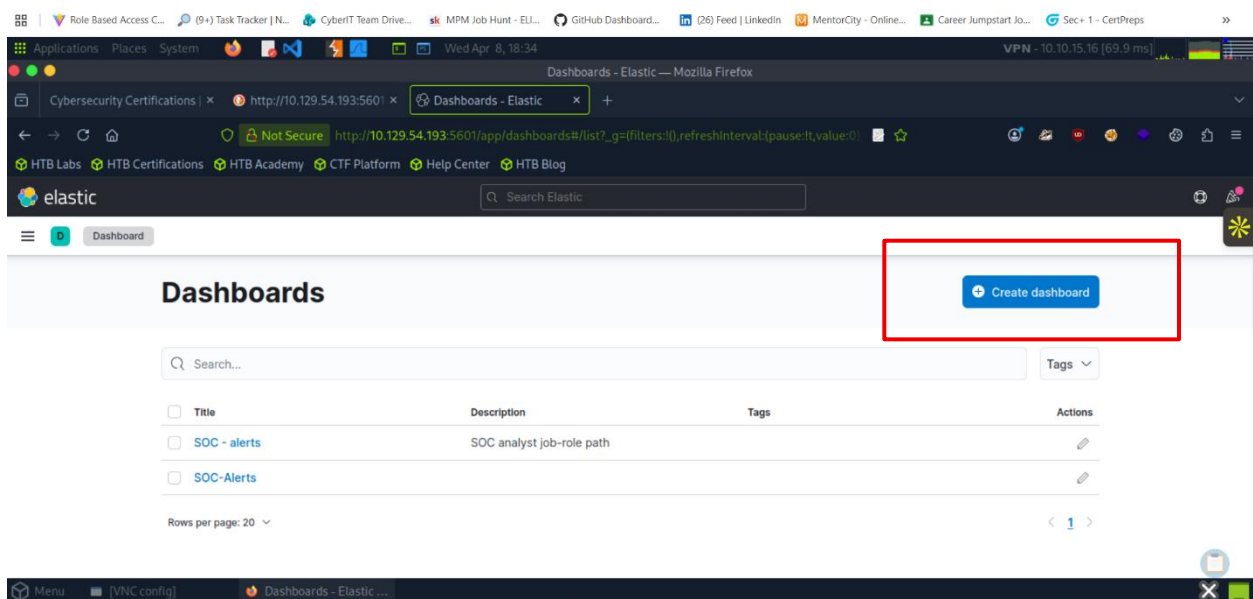
Lab Steps Summary:

1. Dashboard Creation

- Accessed Kibana via target IP

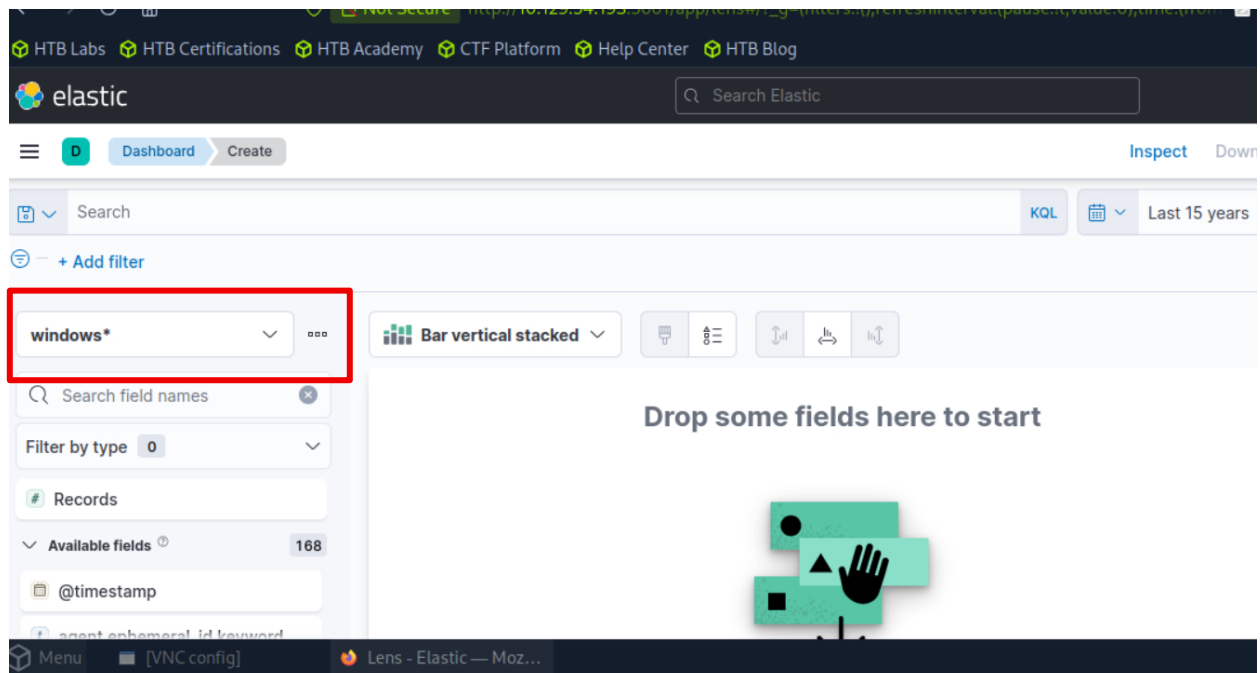
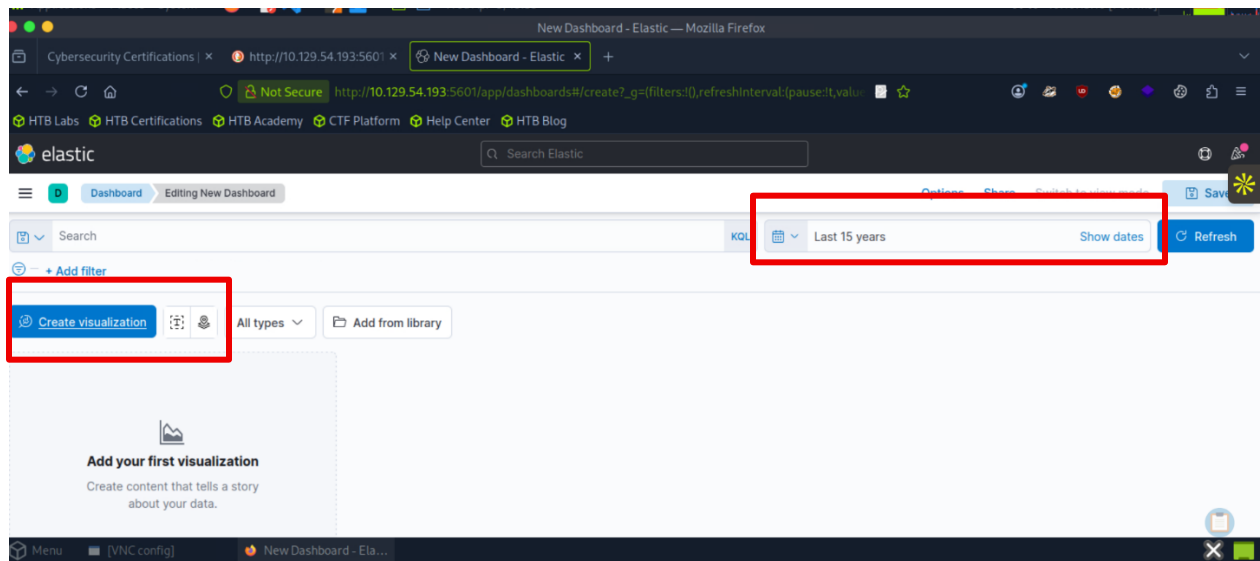


navigate to `http://[Target IP]:5601`, click on the side navigation toggle, and click on "Dashboard"



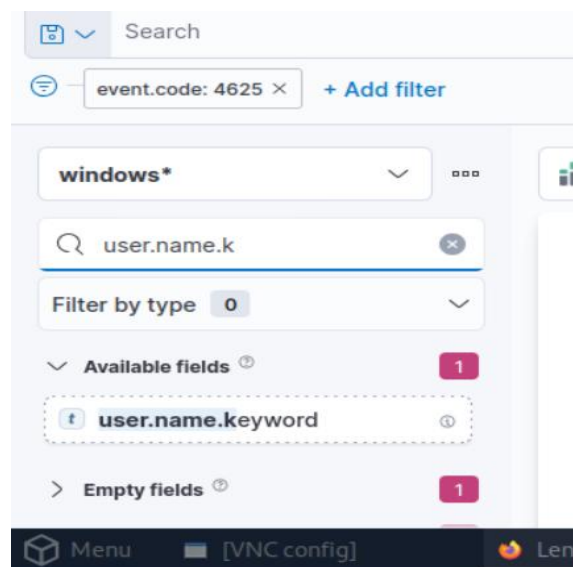
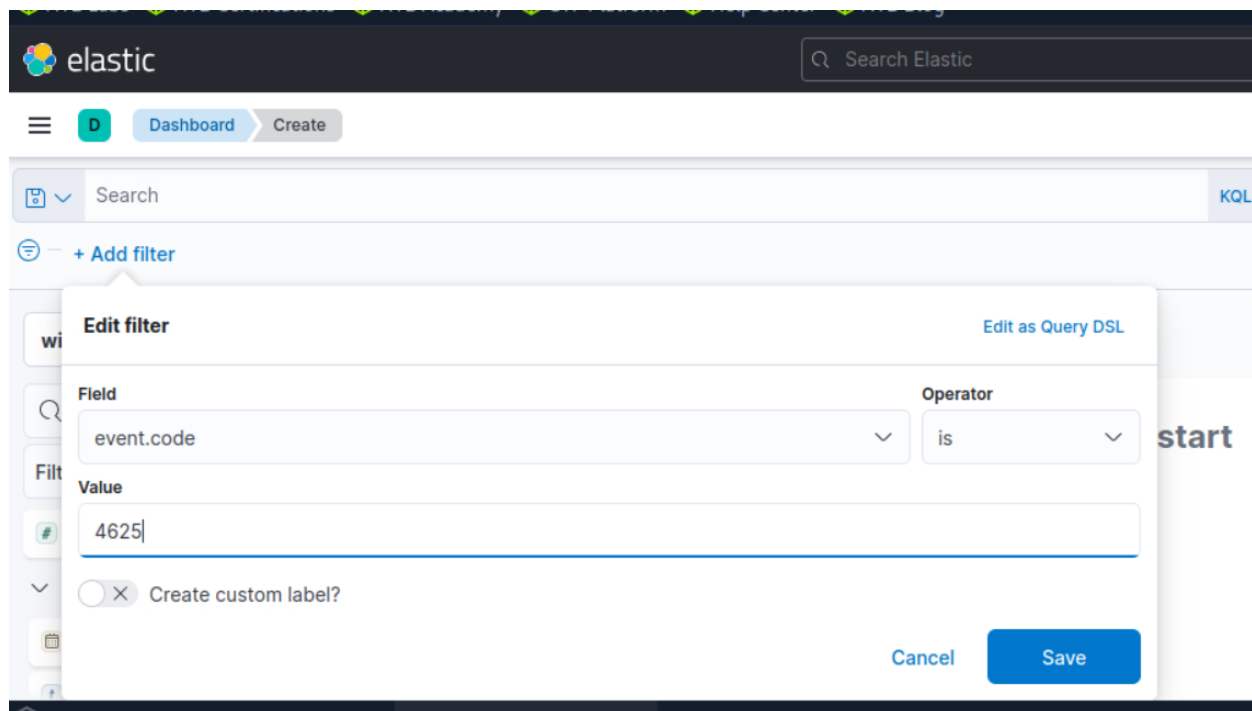
Created a new dashboard and initiated a visualization

Upon initiating the creation of our first visualization, the following new window will appear with various options and settings. Before proceeding with any configuration, it is important for us to first click on the calendar icon to open the time picker. Then, we need to specify the date range as "last 15 years". Finally, we can click on the "Apply" button to apply the specified date range to the data.



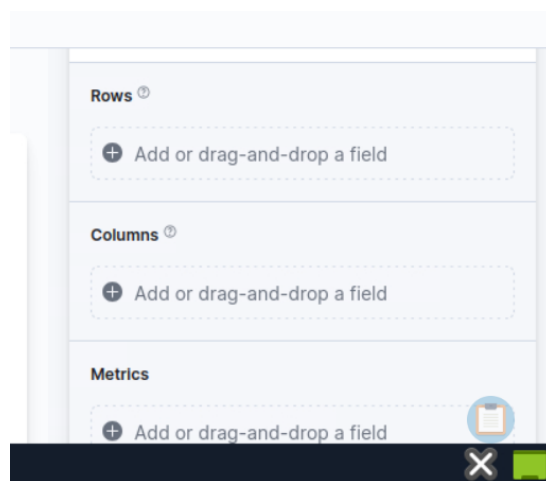
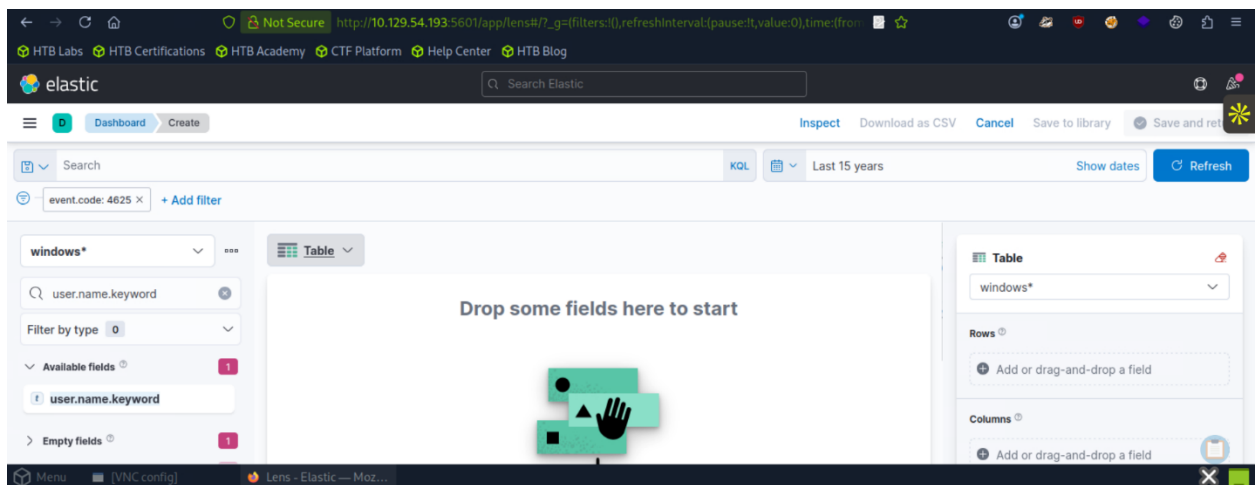
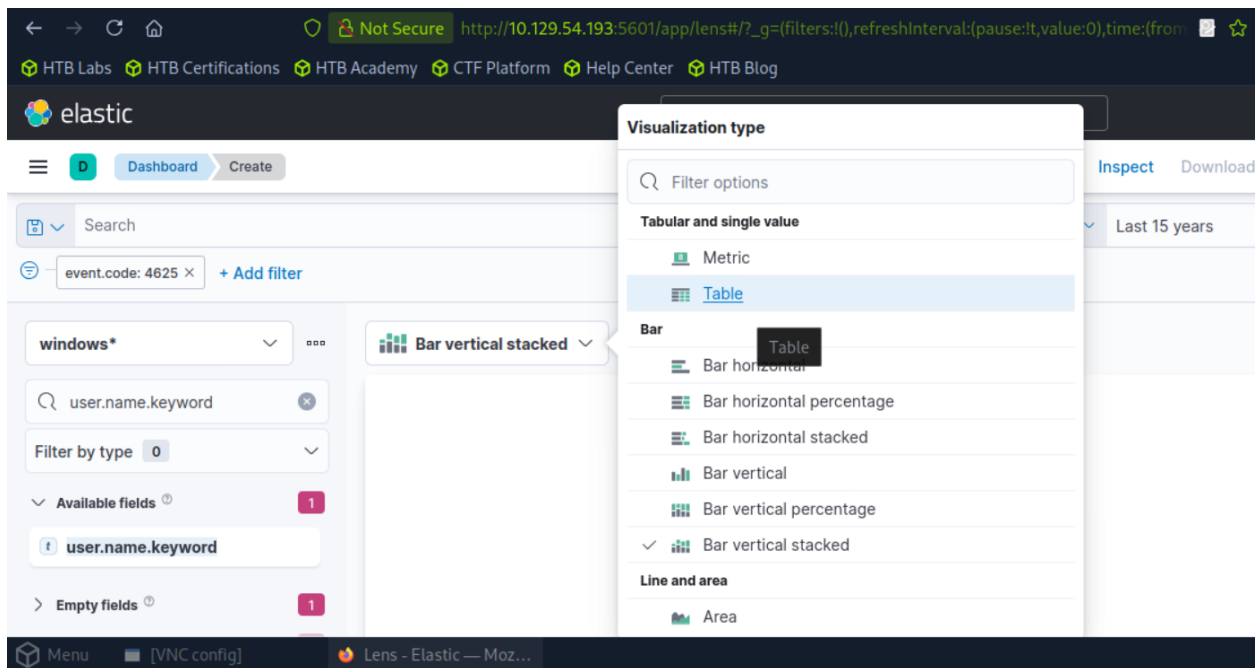
2. Data Configuration

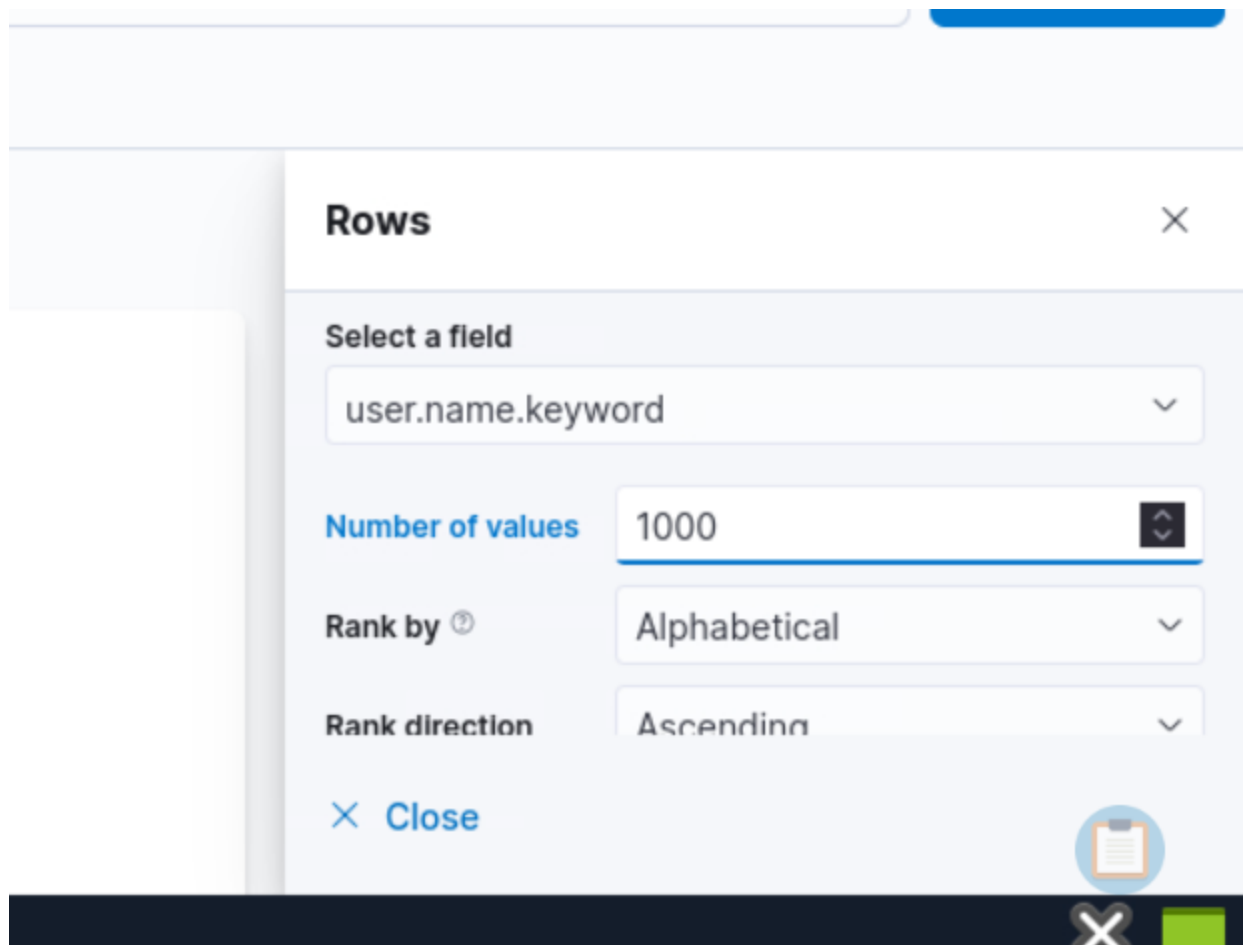
- Selected index pattern: windows*
- Set time range to **Last 15 Years**
- Applied filter: event.code: 4625 (Failed logon attempts)



3. Visualization Setup

- Chose **Table visualization**
- Configured fields:
 - user.name.keyword → Username
 - host.hostname.keyword → Event logged by
 - winlog.logon.type.keyword → Logon Type
- Metric:
 - Count of records (# of logins)





Elastic interface showing visualization type options with 'Bar vertical stacked' selected, including other options like 'Metric' and 'Table'.

For this visualization, let's select the "Table" option. After selecting the "Table", we can proceed to click on the "Rows" option.

This will allow us to choose the specific data elements that we want to include in the table view. Elastic table configuration interface with options to add or drag-and-drop fields for rows, columns, and metrics.

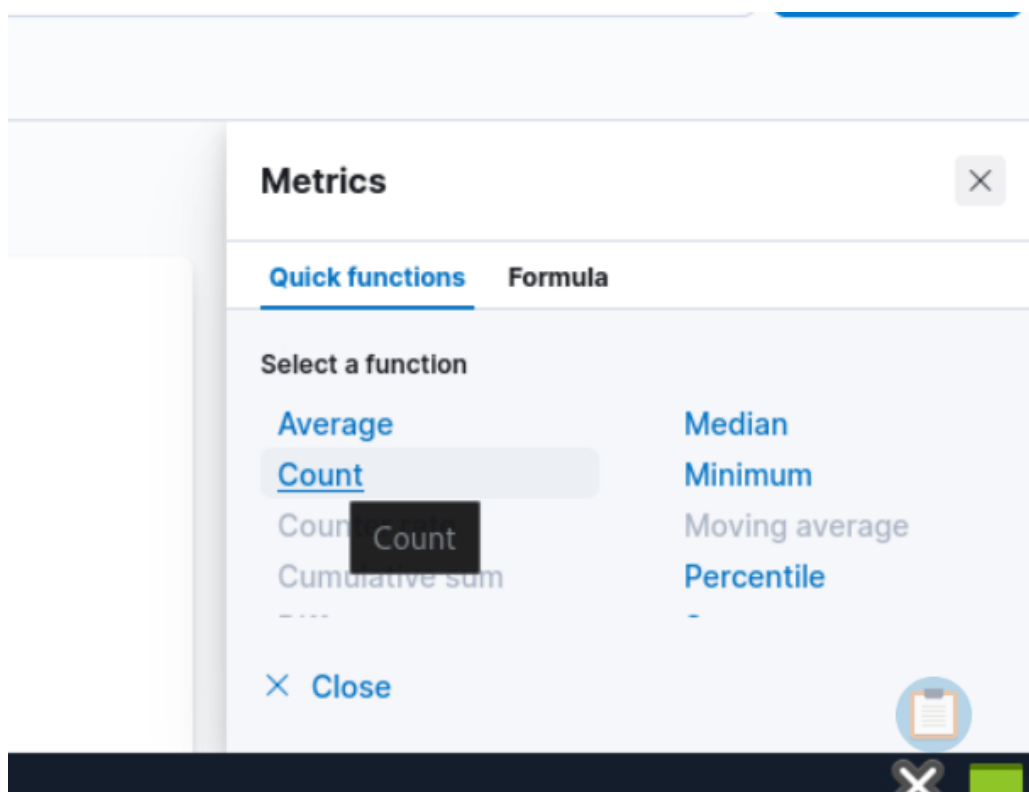
Let's configure the "Rows" settings as follows. Elastic interface for configuring rows, selecting 'user.name.keyword' field, displaying top 1000 values, ranked by count of records in descending order.

Moving forward, let's close the "Rows" window and proceed to enter the "Metrics" configuration. Elastic table configuration showing 'windows*' index pattern, with 'Top values of user.name.keyword' in rows, and options to add fields to columns and metrics.

In the "Metrics" window, let's select "count" as the desired metric. Elastic metrics configuration interface showing quick functions like Average, Count, and Sum, with 'Count' selected. As soon as we select "Count" as the metric, we will observe that the table gets populated with data (assuming that there are events present in the selected data set) Elastic table showing top values of 'user.name.keyword' with counts, and metrics configuration set to 'Count' for records.

One final addition to the table is to include another "Rows" setting to show the machine where the failed logon attempt occurred. To do this, we will select the host.hostname.keyword field, which represents the computer reporting the failed logon attempt.

This will allow us to display the hostname or machine name alongside the count of failed logon attempts, as shown in the image. Elastic table showing top values of 'user.name.keyword' and 'host.hostname.keyword' with record counts, configured in rows. Now we can see three columns in the table.



The screenshot shows the Elastic Dashboard interface. At the top, there's a search bar with the query 'event.code: 4625' and a filter for 'Last 15 years'. Below the search bar, there's a sidebar with 'Available fields' and 'Empty fields'. The main content area displays a table titled 'Top values of user.name.keyword' with the following data:

Top values of user.name.keyword	Count of records
DC1\$	12
EAGLE.LOCAL/ESCACC\$	12
Administrator	4
DESKTOP-DPOESND	4
PAW	4

On the right side, there's a 'Columns' section with 'Add or drag-and-drop a field' and a 'Metrics' section with 'Count of records' and 'Add or drag-and-drop a field'.

4. Refinements

- Renamed columns for better clarity
- Added sorting (descending by login count)
- Excluded specific system/machine accounts

The screenshot shows the Elastic Dashboard interface with a refined table. The table has two columns: 'Top values of user.name.keyword' and 'Top values of host.hostname.keyword', and a 'Count of records' column. The data is as follows:

Top values of user.name.keyword	Top values of host.hostname.keyword	Count of records
DC1\$	DC1	10
DC1\$	DC2	2
EAGLE.LOCAL/ESCACC\$	PKI	12
Administrator	DC1	4
DESKTOP-DPOESND	DC1	4

On the right side, there's a 'Rows' panel with 'Select a field' (host.hostname.keyword), 'Number of values' (1000), and 'Rank by' (Count of records). The 'Close' button is visible at the bottom of the panel.

The screenshot shows the Elastic Dashboard interface with a 'Save dashboard' dialog box open. The dialog box has a 'Title' field with the value 'SOC Alert 1' and a 'Description' field with the value 'Example 1: Failed Logon Attempts (All Users)'. There are 'Cancel' and 'Save' buttons at the bottom of the dialog box. The background shows the dashboard with the same table as in the previous screenshot.

HTB LabsHTB CertificationsHTB AcademyCTF PlatformHelp CenterHTB Blog

elastic

Search Elastic

Dashboard

Editing SOC Aleart 1

Top values of user.name.keywoi	Top values of host.hostname.ke	Count of records
PAW	DC2	4
WIN-OK9BH1BCKSD	DC1	4
WIN-RMMGJA7T9TC	DC1	4
administrator	PAW	3
administrator	DC2	1
bob	WS001	2
sql-svc1	PKI	2
anni	WS001	1
eAdministrator	DC1	1
eagleAdministrator	DC1	1

Menu

[VNC config]

SOC Aleart 1 - Elastic...

Cybersecurity Certifi...

Final Output

The final dashboard displays:

- Username attempting login
- Host machine where the attempt occurred
- Logon type
- Number of failed login attempts